

Circles

Chapter 10 - Hinglish Notes, Tangents, Theorems + Solutions

Ye Chapter 10 (Circles) ka complete guide hai. Pehle red-page ke core basics (circle ke parts, perpendicular, Pythagoras, tangent vs secant), fir chapter ka main content - tangent ke theorems aur tangent ki length. End me solved examples aur 20 practice questions ke step-by-step solutions hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

PART A - Core Basics (Miss Mat Karna)

1. Circle ke Parts

EXAM * Exam me CONSTANTLY aata hai

- Centre = beech ka point.
- Radius (r) = centre se circle tak ki doori.
- Diameter = $2r$ (circle ke aar-paar, centre se hokar).
- Chord = circle pe do points ko jodne wali line.

2. Perpendicular (90 degree)

EXAM * Exam me CONSTANTLY aata hai

- Do lines jo 90 degree par milti hain.
- Symbol: corner par chhota square banate hain.

3. Right Triangle + Pythagoras

EXAM * Exam me CONSTANTLY aata hai

- 90 degree wale triangle me: $\text{hyp}^2 = \text{side1}^2 + \text{side2}^2$.
- Tangent ki length nikaalne me use hota hai.

4. Square / Square Root

- $a^2 = a \times a$. $\text{sqrt}(100) = 10$.
- Tangent ki length nikaalne me sqrt use hota hai.

5. Tangent vs Secant (Is Chapter ka Dil)

EXAM * Exam me CONSTANTLY aata hai

- SECANT = line jo circle ko 2 points par CUT kare.
- TANGENT = line jo circle ko sirf 1 point par TOUCH kare.
- Touch point = 'point of contact'.

6. Number of Tangents

EXAM * Exam me CONSTANTLY aata hai

- Circle ke ANDAR ke point se: 0 tangent.
- Circle PE point se: 1 tangent.
- Circle ke BAHAR ke point se: 2 tangent.

7. Angle Bisector

- Ek line jo kisi angle ko 2 barabar hisson me baant-ti hai.
- Centre aur external point ko jodne wali line tangents ke angle ko bisect karti hai.

PART B - Chapter 10 ka Main Content

8. Tangent to a Circle

EXAM * Exam me CONSTANTLY aata hai

Definition: Wo line jo circle ko sirf ek hi point par touch kare. Us point ko point of contact kehte hain.

9. Theorem 1 - Radius Tangent par Perpendicular

EXAM * Exam me CONSTANTLY aata hai

Point of contact par radius, tangent ke PERPENDICULAR (90 degree) hoti hai.

Yani: OP (radius) $\perp$ tangent at P .

- Isliye OP , tangent ke saath right-angle triangle banata hai.

10. Theorem 2 - Do Tangents Barabar

EXAM * Exam me CONSTANTLY aata hai

Bahar ke ek point se khinchi gayi do tangents ki length BARABAR hoti hai.

Yani: $PA = PB$ (P bahar ka point, A aur B contact points).

- Aur wo tangents centre wali line ke saath barabar angle banati hain.

11. Length of Tangent

EXAM * Exam me CONSTANTLY aata hai

Tangent length = $\sqrt{OP^2 - r^2}$

OP = centre se external point ki doori, r = radius.

- Ye Pythagoras se aata hai (kyunki radius $\perp$ tangent).

12. Extra Important Facts

- Do parallel tangents ke beech ki doori = diameter ($2r$).
- Diameter ke dono sire par tangents aapas me parallel hoti hain.
- Chord ke perpendicular bisector hamesha centre se hokar jaata hai.

Quick Revision

Tangent = 1 point touch; Secant = 2 points cut.
Tangents: andar 0, pe 1, bahar 2.
Radius $\perp$ tangent at point of contact (90 degree).
Bahar ke point se 2 tangents barabar ($PA = PB$).
Tangent length = $\sqrt{OP^2 - r^2}$.

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1: $r=6$, $OP=10$. Tangent ki length?

- Step: Tangent = $\sqrt{OP^2 - r^2}$.
- Step: = $\sqrt{10^2 - 6^2} = \sqrt{100 - 36}$.
- Step: = $\sqrt{64}$.

Answer: Tangent = 8

Example 2: $r=5$, tangent length=12. OP (centre se doori)?

- Step: $OP^2 = r^2 + \text{tangent}^2 = 25 + 144$.
- Step: $OP = \sqrt{169}$.

Answer: $OP = 13$

Example 3: Bahar ke point se kitni tangents?

- Step: Rule: bahar ke point se 2 tangents.

Answer: 2

Example 4: $PA = 7$ (ek tangent). Doosri tangent PB?

- Step: Bahar ke point se do tangents barabar.
- Step: $PB = PA$.

Answer: $PB = 7$

Example 5: Radius aur tangent ke beech ka angle?

- Step: Point of contact par radius $\perp$ tangent.

Answer: 90 degree

Example 6: $r=8$, $OP=17$. Tangent length?

- Step: $\sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225}$.

Answer: 15

Example 7: Circle pe point se kitni tangent?

- Step: Circle PE point se exactly 1 tangent.

Answer: 1

Example 8: Andar ke point se kitni tangent?

- Step: Andar ke point se 0 tangent (line cut karegi).

Answer: 0

Example 9: Diameter = 14. Do parallel tangents ke beech doori?

- Step: Parallel tangents ke beech doori = diameter.

Answer: 14

Example 10: $r=9$, $OP=15$. Tangent length?

- Step: $\sqrt{15^2 - 9^2} = \sqrt{225 - 81} = \sqrt{144}$.

Answer: 12

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

- Q1. Radius aur diameter me kya relation hai?
- Q2. Chord kya hota hai?
- Q3. Tangent kya hota hai?
- Q4. Secant kya hota hai?
- Q5. Bahar ke point se kitni tangents khinch sakte hain?
- Q6. Circle pe point se kitni tangents?
- Q7. Andar ke point se kitni tangents?
- Q8. Point of contact par radius aur tangent ke beech angle?
- Q9. Tangent ki length ka formula kya hai?
- Q10. $r=3$, $OP=5 \rightarrow$ tangent length?
- Q11. $r=8$, tangent=6 $\rightarrow$ OP?
- Q12. Bahar ke point se do tangents ke beech kya relation?
- Q13. Point of contact kise kehte hain?
- Q14. Do parallel tangents ke beech doori kiske barabar hoti hai?
- Q15. $r=12$, $OP=13 \rightarrow$ tangent length?
- Q16. Tangent length formula kis theorem se aata hai?
- Q17. $PA=9$ ek tangent ho to doosri PB?
- Q18. Diameter 10 ho to radius?
- Q19. Secant circle ko kitne points par cut karti hai?
- Q20. Centre se chord par dali gayi perpendicular chord ko kya karti hai?

Step-by-Step Solutions

- Q1. Radius aur diameter relation?
 - Step: Diameter centre se hokar dono taraf radius.**Answer:** $\text{Diameter} = 2r$
- Q2. Chord kya hota hai?
 - Step: Circle pe 2 points jodne wali line.**Answer:** Do points jodne wali line
- Q3. Tangent kya hota hai?
 - Step: Circle ko 1 point par touch.**Answer:** 1 point touch karne wali line
- Q4. Secant kya hota hai?
 - Step: Circle ko 2 points par cut.**Answer:** 2 points par cut karne wali line
- Q5. Bahar ke point se tangents?
 - Step: Rule.**Answer:** 2

Q6. Circle pe point se tangents?

- Step: Rule.

Answer:1

Q7. Andar ke point se tangents?

- Step: Line cut karegi.

Answer:0

Q8. Radius-tangent angle?

- Step: Theorem 1.

Answer:90 degree

Q9. Tangent length formula?

- Step: Pythagoras se.

Answer: $\sqrt{OP^2 - r^2}$

Q10. $r=3$, $OP=5 \rightarrow$ tangent?

- Step: $\sqrt{25 - 9} = \sqrt{16}$.

Answer:4

Q11. $r=8$, tangent=6 $\rightarrow$ OP?

- Step: $OP = \sqrt{64 + 36} = \sqrt{100}$.

Answer:10

Q12. Do tangents (bahar se) relation?

- Step: Theorem 2.

Answer:Barabar ($PA = PB$)

Q13. Point of contact?

- Step: Jahan tangent circle ko touch kare.

Answer:Touch karne wala point

Q14. Parallel tangents ke beech doori?

- Step: Diameter ke barabar.

Answer:Diameter ($2r$)

Q15. $r=12$, $OP=13 \rightarrow$ tangent?

- Step: $\sqrt{169 - 144} = \sqrt{25}$.

Answer:5

Q16. Tangent formula kis theorem se?

- Step: Radius $\perp$ tangent $\rightarrow$ Pythagoras.

Answer:Pythagoras theorem

Q17. $PA=9 \rightarrow$ PB?

- Step: Do tangents barabar.

Answer:9

Q18. Diameter 10 $\rightarrow$ radius?

- Step: $r = \text{diameter}/2$.

Answer:5

Q19. Secant kitne points cut?

- Step: Do points par.

Answer:2

Q20. Centre se chord par perpendicular?

- Step: Chord ko do barabar hisson me baant-ti hai.

Answer:Chord ko bisect karti hai